Jarvis Features

The voice assistant of the AI + IoT Smart Home Security System. All of it runs on free software on the team laptop, so it adds 0 TL to the budget.

What Jarvis is

Jarvis is the voice of the house. You say its name, speak normally, and it listens, understands, then either does something (switches the light, locks the door) or answers (temperature, who is home). It can also speak up on its own when something important happens.

How it works, in four steps: (1) it waits for the wake word "Jarvis"; (2) free speech-to-text (Whisper) turns your voice into words; (3) an AI language model works out what you mean; (4) it sends the command to the right device, or reads the answer from the house data, and replies in a spoken voice (Piper).

1. Control commands (it does something)

You say	What happens	Uses
"Turn the light on / off"	Switches the lamp through a relay.	Relay, ESP32
"Turn on the grow light"	Switches the plant light.	Relay, grow light
"Water the plants"	Runs the pump for a set time.	Pump, soil sensors
"Turn on the fan / humidifier"	Starts the bathroom-corner fan or mist.	Relay, fan, humidifier
"Lock the door"	Engages the solenoid lock.	Solenoid lock
"Arm / disarm the security system"	Turns intruder alerts on or off.	Camera AI
"Silence the alarm"	Stops the buzzer.	Buzzer

2. Questions (it reads and answers)

You say	What happens	Uses
"What's the temperature and humidity?"	Reads the sensor and tells you.	BME680
"Is there a mold risk?"	Says if the bathroom corner is in the danger zone.	BME680
"How are the plants?"	Reports soil moisture and the last watering.	Soil sensors
"Is the door locked?"	Gives the lock state.	Lock, fingerprint
"Who's home?"	Lists the people face recognition has seen.	Camera, face ID
"Is anyone at the door?"	Checks the camera for a person.	ESP32-CAM, YOLO
"Any alerts today?"	Reads out the event log.	Server database
"Is everything okay?"	Gives a short status summary of the whole house.	All sensors

3. Conversation

You say	What happens	Uses
Natural phrasing	"It feels stuffy in here" leads to a humidity answer; you do not have to memorise exact phrases.	Language model
Follow-ups	"Do it again" or "and the plants?" works, because it remembers the conversation.	Language model
General questions	Time, date, simple facts, unit conversions.	Language model
English and Turkish	Whisper understands both languages.	Whisper

4. Jarvis speaks up on its own

You say	What happens	Uses
Spoken alerts	"Warning: humidity has been high for 10 minutes." "Unknown person at the front door."	All alerts
Greeting by face	When face recognition sees someone: "Welcome home, Sara."	Camera, face ID
Proactive suggestions	"Humidity is rising. Shall I turn on the fan?"	BME680, relay
Intruder response	Alarm sounds, light comes on, Jarvis announces the owner was notified.	Camera, buzzer
Fire response	Jarvis announces the fire and where, sounds the alarm and releases the door lock so people can leave.	Flame / gas sensors, lock

5. Scenes and routines (one phrase, many actions)

You say	What happens	Uses
"Goodnight"	Locks the door, arms security, turns off the light.	Lock, relay
"I'm leaving"	Locks up, arms security, starts away mode.	Lock, camera
"I'm home"	Disarms, turns on the light, greets you by name.	Face ID
Away mode	Switches the light on and off at random times so the house looks lived in.	Relay
Water schedule	"Water the plants every morning at 8."	Pump
Reminders and timers	"Remind me to check the plants at 6" or "Set a 10 minute timer."	Laptop

6. Smart extras

You say	What happens	Uses
Visitor description	"Who's at the door?" takes a snapshot and describes it, e.g. a person in a blue jacket.	ESP32-CAM, vision AI
Daily report	"What happened today?" summarises entries, alerts and plant watering.	Server database
Plant advice	"Why are the leaves drooping?" answers from soil moisture and light data.	Soil sensors
Guest mode	"Let my friend in tomorrow" creates a temporary door code that expires.	Keypad, lock
Add a new face	Jarvis guides you through enrolling a new family member's face.	Face ID
Self-check	"Run a system check" reports which devices are online and which are not.	All boards
Do not disturb	"Quiet mode" mutes spoken alerts at night but keeps the alarm for emergencies.	Laptop
Dashboard transcript	A chat window on the web page shows what Jarvis heard and did.	Dashboard

Safety rules

Unlocking by voice is risky. A recording of your voice, or anyone nearby, could open the door. Jarvis should not unlock by voice alone. Require a spoken PIN, or the fingerprint or face check first. Locking, arming and switching lights are safe by voice.

Confirm out loud. Jarvis says what it did ("Light on") so you know it understood.
Wake word only. It listens for "Jarvis" and does not record you all the time.
Fire exception. In a fire, releasing the lock is a deliberate safety feature, triggered by the sensors, not by voice.

Cost

Whisper, Piper, the language model (a small local one, or a free API tier), the wake-word tool and the dashboard are all free. The only hardware needed is a microphone and speaker, which the laptop already has. Total extra cost: 0 TL.